

Optimization of Cantilever Beams with Multiple Flexoelectric Actuators Based on Newton-Raphson Iteration

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Abstract This study delves into the complexity of optimizing multiple actuators on a cantilever beam, focusing on the flexoelectric effect caused by the non-uniform electric field generated by an Atomic Force Microscope (AFM) probe. Multiple actuators control has significant research value in enhancing the flexoelectric effect, greatly alleviating stress concentration and achieving precise vibration control. The current challenge in multiple flexoelectric actuators research is multi-objective optimization, addressed here using the Newton-Raphson iterative method, known for its robustness in the convex function domain, as an optimization framework. By analyzing structural parameters and flexoelectric actuator parameters, significant influencing factors are selected to form the vector space, determining actuator positions and driving voltages. These variables constitute the optimization space and are incorporated into the Newton-Raphson general iterative equation to derive the iteration matrix, which is computed using MATLAB. Case studies confirm that the Newton-Raphson method effectively identifies optimal actuator positions and driving voltages at different modes without external force, significantly improving flexoelectric control efficiency. Additionally, it quickly stabilizes vibrations at different modes under external force. However, the study has limitations, as the Newton-Raphson method cannot effectively solve non-convex function optimization in linear space. This research advances the understanding of multiple actuators optimization control structure dynamics and promotes the development of more effective engineering solutions, particularly in achieving more precise actuation and control in the field of micro- and nano-structure engineering.

Key words: Flexoelectric effect, Vibration control, Newton-Raphson iteration, Cantilever beams

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1. Introduction

The cantilever beam is one of the most common structural models in engineering applications, such as fixed aircraft wings[1], robotic arms[2], instrument equipment[3], and bridges[4]. In recent decades, precise actuation and vibration control of cantilever beams have been widely studied. In theoretical research, modal expansion is an effective tool for in-depth investigation of the dynamic characteristics

of cantilever beams[5, 6]. Various smart materials have made progress as sensors and actuators. Among them, piezoelectric materials have been widely used in engineering applications as sensors and actuators based on the piezoelectric effect, achieving precise actuation or vibration control[7-10]. Other smart materials include shape memory materials[11], electro/magnetostrictive materials[12], photostrictive materials[13], and superconductor materials[14], all of which can be used in vibration control. In recent years, smart materials based on the reverse flexoelectric effect have become a research hotspot as an alternative to piezoelectric materials[15]. However, compared to piezoelectric materials, flexoelectric materials tend to exhibit more severe stress concentration. To address this issue, deploying multiple flexoelectric actuators is an effective solution.

Research by Fan et al.[16-18] evaluated flexoelectric beams actuated by the reverse flexoelectric effect and found that the control effect on the beam increases with thinner linear electrode radii and thinner flexoelectric beams. Subsequently, the study analyzed the case where the Young's modulus and mass density of the flexoelectric layer were higher than the elastic layer, showing that with an increase in the flexoelectric layer's thickness, the top displacement of the laminated beam decreased rapidly. Through case studies, optimal control positions for each vibration mode were found to depend on the flexoelectric layer properties. The study introduced a displacement-based feedback control to avoid overactuation caused by open-loop control. Zhang et al.[19] used velocity feedback to adjust the system's damping ratio, providing an optimal choice for the closed-loop control design of a flexoelectric-actuated parabolic shell structure. Deng et al.[20] used PbLaZrTi (PLZT) as a photovoltaic generator to actuate a flexoelectric actuator, studying static deflection control and dynamic vibration control through Lyapunov control. Zhang's et al. research results suggested that in a cantilever beam, the control effect of

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flexoelectric fibers is more effective than piezoelectric fibers[21]. Wang et al.[22] studied an array of flexoelectric nanobeam energy harvesters and provided important information for the design of flexoelectric collector arrays. Barati[23] extensively studied the nonlinear vibration of nanobeams, considering the effects of temperature change, elastic foundation, surface coefficients, and nanobeam thickness. Xia et al.[24] revealed the temperature-modulated piezo-flexoelectric effect in ferroelectric Ba_{0.7}Sr_{0.3}TiO₃ materials, providing a convenient route for scavenging and sensing vibration energy.

The Newton-Raphson method, known as Nonlinear programming algorithm, is a numerical computation method for finding the roots of a function. In recent years, nonlinear programming algorithms and finite element method have been widely applied[25-30]. The Newton-Raphson iteration method has also been widely used in research across different fields through MATLAB programming[31]. Kusekwa and Mashauri [32] used this algorithm to study power system state estimation, optimal power flow, and security-constrained optimal power flow in the Tanzanian power network. Chatterjee et al.[33] analyzed and compared the convergence characteristics of different algorithms in power flow, concluding that the Newton-Raphson algorithm exhibits quadratic convergence. Gamberi et al.[34] used this algorithm to model multi-zone hydronic heating systems in buildings more conveniently, improving the efficiency of system analysis. Hilderbrandt et al.[35], in the field of molecular mechanics, utilized the Newton-Raphson algorithm for geometric optimization, normal coordinate analysis, thermodynamic calculations, and the synthesis of electron diffraction radial distribution curves.

In this study, an analysis of the influence of a single flexoelectric patch on the cantilever beam's tip displacement under different parameters was first conducted. Based on these results, parameters with

significant effects were selected, and the Newton-Raphson algorithm was then used to analyze the optimal arrangement of multiple actuators to achieve optimal driving and control effects. Finally, a negative feedback closed-loop adjustment algorithm for free decay amplitude based on the Newton-Raphson algorithm was proposed. The computational results indicate that compared to free decay vibrations, the tip displacement of the cantilever beam can be rapidly stabilized with this algorithm.

2. Physical Model

As previously discussed, the primary objective in employing multiple flexoelectric actuators is to mitigate stress concentration issues inherent in flexoelectric actuation. The schematic representation of a cantilever beam mode equipped with multiple flexoelectric actuators is depicted in **Fig.1** and **Fig.2**. In order to induce the converse flexoelectric effect, it is imperative to create an inhomogeneous electric field within the flexoelectric patches, from which control forces are subsequently derived via the flexoelectric actuators.

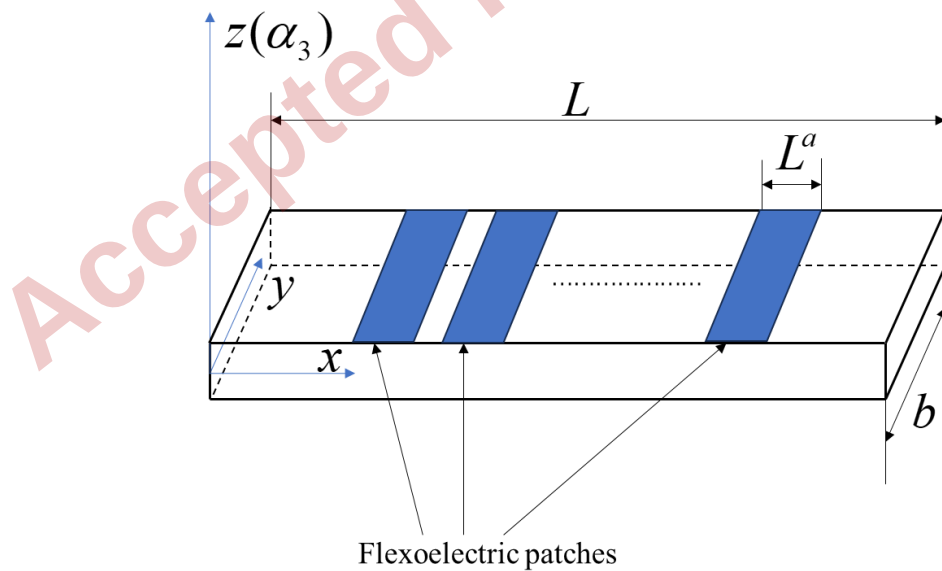


Fig. 1 Cantilever beam with multiple flexoelectric actuators

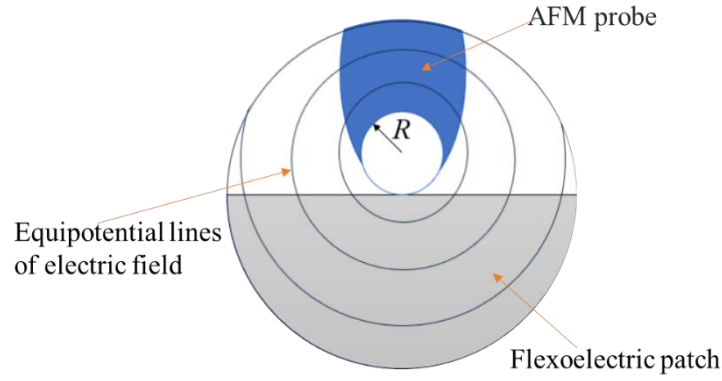


Fig. 2 Schematic diagram of the electric field generated by an AFM probe

Within **Fig.1**, the Cartesian coordinates x and z delineate the longitudinal and transverse directions, respectively, with L signifying the length of the cantilever beam, while b and h denote the beam's width and thickness, respectively. Regarding the actuators, the dimensional attributes of the Atomic Force Microscope (AFM) probes are characterized by the probe radius R , whereas L_a and h_a represent the thickness and length of each patch, correspondingly. For analytical simplification, it is assumed that all patches share identical dimensions. The positional notation for the i th actuator on the beam is denoted by x_i , where i takes values in the range of 1, 2, 3, and so forth, up to n . This notation signifies the distance measured from the fixed terminus of the beam to the midpoint of the patch.

Furthermore, it is assumed that the probe radius R , patch thickness h_a , and beam thickness h adhere to the following relationship, namely, $R \ll h_a \ll h$. This relation underscores the pronounced disparity in dimensions, where the probe radius R is significantly smaller than the patch thickness, and the patch thickness is substantially smaller than the beam thickness, an assumption that underpins the ensuing analysis.

2.1 Vibration equation of cantilever beams

An applied mechanical force, denoted as F , exerts its influence at the unbound terminus of the beam, thereby initiating a static deformation within the structural framework. This analysis presupposes the

commencement of the beam's motion at $t = 0$. Taking into account the absence of external mechanical forces, considering only the effect of flexoelectric actuation on the vibration of the cantilever beam, the dynamic equation of the cantilever beam is as follows:

$$\rho A \ddot{u}_3 + YI \frac{\partial^4 u_3}{\partial x^4} = b \frac{\partial^2 M_{xx}^a}{\partial x^2} \quad (1).$$

In the Eq.(1), where A represents the cross-sectional area of the beam, Y denotes the Young's modulus of the material, and I represents the moment of inertia of the cross-section, defined as $I = bh^3 / 12$.

M_{xx}^a represents the bending moment generated by flexoelectricity.

The transverse displacement response of the beam is theoretically expressed through the rigorous application of the modal expansion method.

$$u_{3k}(x, t) = \sum_{i=1}^{\infty} \eta_k(t) U_{3k}(x) \quad (2)$$

where k is the mode number; η_k is the k th modal participation factor;

By applying the principle of modal orthogonality for cantilever beams, we can express the equation for the mode participation factor (or the k th modal equation) as follows:

$$\ddot{\eta}_k + 2\zeta_k \omega_k \dot{\eta}_k + \omega_k^2 \eta_k = \hat{F}_k(t) \quad (3).$$

in which, ω_k denotes the inherent natural frequency of the system, while ζ_k represents the modal damping ratio. The latter can be precisely defined as a function of the equivalent damping constant c , which is $\zeta_k = c / (2\rho h \omega_k)$.

Notably, $\hat{F}_k(t)$ designates the modal excitation force associated with the k th mode, and it is conspicuously absent in the unconstrained response of the cantilever beam when $t > 0$. U_{3k} is the cantilever beam's k th mode shape function, which can be written as

$$U_{3k}(x) = C_k \left[\cosh \lambda_k x - \cos \lambda_k x - \frac{\cosh \lambda_k L + \cos \lambda_k L}{\sinh \lambda_k L + \sin \lambda_k L} (\sinh \lambda_k x - \sin \lambda_k x) \right] \quad (4).$$

In this context, C_k represents the modal amplitude associated with the mode; $\lambda_k L$ signifies the root of the characteristic equation $1 + \cosh(\lambda L) \cos(\lambda L) = 0$; and the initial three roots are $\lambda_k L = 1.875, 4.694, 7.855 (k = 1, 2, 3)$. The natural frequency of the cantilever beam structure pair is

$$\omega_k = \lambda_k^2 \sqrt{YI / \rho A}.$$

Eq.(3) can be solved by using Laplace transformation:

$$\begin{aligned} \eta_k(t) = & e^{-\xi_k \omega_k t} \left\{ \eta_k(0) \cos(\omega_{dk} t) + [\dot{\eta}_k(0) \xi_k \omega_k + \dot{\eta}_k(0)] \frac{\sin(\omega_{dk} t)}{\omega_{dk}} \right\} \\ & + \frac{1}{\omega_{dk}} \int_0^t \hat{F}_k(\tau) e^{-\xi_k \omega_k (t-\tau)} \sin[\omega_{dk} (t-\tau)] d\tau \end{aligned} \quad (5).$$

in which $\omega_{dk} = \omega_k \sqrt{1 - \xi_k^2}$, and $\eta_k(0)$ can be solved from the static function:

$$\omega_k^2 \eta_k(0) = \hat{F}_k(0) \quad (6).$$

in which the mechanical force F induces the modal force $\hat{F}_k(0)$, which can be expressed as:

$$\hat{F}_k(0) = \frac{1}{\rho h N_k} \int_0^L F \delta(x-L) U_{3k}(x) dx = \frac{F U_{3k}(L)}{\rho h N_k} \quad (7).$$

N_k is defined as $N_k = \int_0^L U_{3k}^2 dx$. Substituting Eq.(7) into Eq.(6) yields a new equation:

$$\eta_k(0) = \frac{\hat{F}_k}{\omega_k^2} = \frac{F U_{3k}(L)}{\omega_k^2 \rho h N_k} \quad (8).$$

Substituting Eq.(8) into Eq.(5), noting that $\hat{F}_k(t) = 0$ and $\dot{\eta}_k(0) = 0$, (when $t > 0$), the displacement of the beam can be expressed as:

$$\begin{aligned} u_3(x, t) = & \sum_{k=1}^{\infty} \eta_k(t) U_{3k}(x) \\ = & \sum_{k=1}^{\infty} e^{-\xi_k \omega_k t} \frac{F U_{3k}(L) U_{3k}(x)}{\omega_k^2 \rho h N_k} \left[\cos(\omega_{dk} t) + \frac{\xi_k}{\sqrt{1 - \xi_k^2}} \sin(\omega_{dk} t) \right] \\ = & \sum_{k=1}^{\infty} e^{-\xi_k \omega_k t} \frac{F U_{3k}(L) U_{3k}(x)}{\omega_k^2 \rho h N_k} \frac{e^{j(\omega_{dk} t - \varphi_k)}}{\sqrt{1 - \xi_k^2}} \end{aligned} \quad (9).$$

in which the phase difference φ_k can be expressed as

$$\varphi_k = \arctan \frac{\xi_k}{\sqrt{1 - \xi_k^2}} \quad (10).$$

2.2 The flexoelectric actuation mechanism under probe excitation

The flexoelectric actuation coefficient of the cantilever beam structure is

$$L_3^a = -\frac{\partial^2 M_{xx}^a}{\partial x^2} \quad (11).$$

Taking into consideration the modal force formula induced by the flexoelectric effect of a flexoelectric cantilever beam:

$$\hat{F}_{mn}^a = \frac{1}{\rho h N_{mn}} \int \int L_3^a (E_x, E_y, E_z) U_{3mn}(x) dx dy \quad (12).$$

Combining Eq.(5) and Eq.(6), the actuation force generated by the flexoelectric actuator on the cantilever beam structure is obtained as follows:

$$\hat{F}_k^a = \frac{1}{\rho h N_k} \int_{x_i-L/2}^{x_i+L/2} L_3^a U_{3k} dx = \frac{1}{\rho h N_k} \int_{x_i-L/2}^{x_i+L/2} -\frac{\partial^2 M_{xx}^a}{\partial x^2} U_{3k} dx \quad (13).$$

When the modal force is in harmonic form, Eq.(3) can be solved for an analytical solution. Assuming the excitation voltage of the flexoelectric actuator is in harmonic form with a frequency of ω_i , the harmonic response induced by the excitation of the flexoelectric effect, in which $\eta_{mn} = \eta_{mn}^a e^{j\omega_i t}$, can be expressed as the modal participation factor:

$$\begin{aligned} \eta_k &= \frac{\hat{F}_k^a}{(\omega_k^2 - \omega_i^2) + 2j\zeta_k \omega_k \omega} e^{j\omega_i t} \\ &= \frac{\hat{F}_k^a}{\omega_k^2 \sqrt{[1 - (\omega_i / \omega_k)^2]^2 + 4\zeta_k^2 (\omega_i / \omega_{mn})^2}} e^{j(\omega_i t - \varphi_k)} \end{aligned} \quad (14).$$

in which the phase difference φ_k can be expressed as

$$\varphi_k = \arctan \frac{2\zeta_k (\omega_i / \omega_k)}{1 - (\omega_i / \omega_k)^2} \quad (15).$$

According to the reverse flexoelectric effect, when an electric voltage is applied between the electrodes of the flexoelectric patch and the AFM probe, the electric field gradient generated by the AFM probe will induce stress within the flexoelectric patch, which can be utilized to control the vibration of the

beam. When a voltage of ϕ_i^a is applied to the probe and acts on the i th actuator, the electric potential within the i th flexoelectric patch can be approximately expressed as follows:

$$\phi_i = \frac{R(R+2h^a)\phi_i^a}{2h^a} \left[\frac{1}{\sqrt{(x-x_i)^2 + (z-h^a-R-h/2)^2}} - \frac{1}{\sqrt{(x-x_i)^2 + (z+h^a+R-h/2)^2}} \right] \quad (16).$$

According to Maxwell's equations, the transverse direction electric field between the flexoelectric patch and the AFM probe can be expressed as

$$E_{zi} = -\frac{\partial \phi}{\partial z} = \frac{R(R+2h^a)\phi_i^a}{4h^a} \left[\frac{2z-2h^a-2R-h}{\left[(x-x_i)^2 + (z-h^a-R-h/2)^2 \right]^{\frac{3}{2}}} - \frac{2z+2h^a+2R-h}{\left[(x-x_i)^2 + (z+h^a+R-h/2)^2 \right]^{\frac{3}{2}}} \right] \quad (17).$$

Therefore, the stress formula generated by the flexoelectric actuator through the reverse flexoelectric effect can be expressed as:

$$T_{xxi}^a = \pi_{12} \frac{\partial E_{zi}}{\partial z} = -\frac{\pi_{12}R(R+2h^a)\phi_i^a}{2h^a} \left[\frac{3(2z-2h^a-2R-h)^2}{4\left[(x-x_i)^2 + (z-h^a-R-h/2)^2 \right]^{\frac{5}{2}}} - \frac{1}{\left[(x-x_i)^2 + (z-h^a-R-h/2)^2 \right]^{\frac{3}{2}}} - \frac{3(2z+2h^a+2R-h)^2}{4\left[(x-x_i)^2 + (z+h^a+R-h/2)^2 \right]^{\frac{5}{2}}} - \frac{1}{\left[(x-x_i)^2 + (z+h^a+R-h/2)^2 \right]^{\frac{3}{2}}} \right] \quad (18).$$

In the equation, T_{xxi} represents the normal stress in the x direction. The thin film force of the flexoelectric patch can be calculated by integrating the stress in the thickness direction of the flexoelectric layer:

$$N_{xxi}^a = \int_{h/2}^{h/2+h^a} T_{xxi}^a dz = -\frac{\pi_{12} R (R+2h^a) \phi_i^a}{2h^a} \left[\frac{R}{\left[(x-x_i)^2 + R^2 \right]^{\frac{3}{2}}} + \frac{2h^a + R}{\left[(x-x_i)^2 + (2h^a + R)^2 \right]^{\frac{3}{2}}} - \frac{2h^a + 2R}{\left[(x-x_i)^2 + (h^a + R)^2 \right]^{\frac{3}{2}}} \right] \quad (19).$$

So, the force generated by the flexoelectric effect on a unit width of the elastic beam results in a moment:

$$M_{xxi}^a = \frac{(h^a + h) N_{xxi}^a}{2} = \frac{\pi_{12} R (R+2h^a) (h^a + h) \phi_i^a}{4h^a} \left[-\frac{R}{\left[(x-x_i)^2 + R^2 \right]^{\frac{3}{2}}} - \frac{2h^a + R}{\left[(x-x_i)^2 + (2h^a + R)^2 \right]^{\frac{3}{2}}} + \frac{2h^a + 2R}{\left[(x-x_i)^2 + (h^a + R)^2 \right]^{\frac{3}{2}}} \right] \quad (20).$$

By substituting the control moments generated by multiple actuators into the modal force expression,

assuming there are p flexoelectric actuators, the total modal force can be obtained:

$$\begin{aligned} \hat{F}_k^a &= \sum_{i=1}^p \hat{F}_{ki}^a = \sum_{i=1}^p \frac{1}{\rho h N_k} \int_{x_i-L_a/2}^{x_i+L_a/2} -\frac{\partial^2 M_{xx}^a}{\partial x^2} U_{3k} dx \\ &= \sum_{i=1}^p -\frac{3\pi_{12} R (R+2h^a) (h+h^a) \phi_i^a}{4\rho h N_k h^a} \int_{x_i-L_a/2}^{x_i+L_a/2} \left[\frac{5R^3}{\left[(x-x_i)^2 + R^2 \right]^{\frac{7}{2}}} - \frac{4R}{\left[(x-x_i)^2 + R^2 \right]^{\frac{5}{2}}} + \frac{5(2h^a + R)^3}{\left[(x-x_i)^2 + (2h^a + R)^2 \right]^{\frac{7}{2}}} - \frac{4(2h^a + R)}{\left[(x-x_i)^2 + (2h^a + R)^2 \right]^{\frac{5}{2}}} \right. \\ &\quad \left. - \frac{10(h^a + R)^3}{\left[(x-x_i)^2 + (h^a + R)^2 \right]^{\frac{7}{2}}} + \frac{8(h^a + R)}{\left[(x-x_i)^2 + (h^a + R)^2 \right]^{\frac{5}{2}}} \right] U_{3k} dx \end{aligned} \quad (21).$$

Substituting the total modal force into the modal participation factor Eq.(8), the displacement expression

of the cantilever beam can be derived from Eq.(2):

$$\begin{aligned}
u_{3k}(x, t) &= \sum_{i=1}^p \sum_{k=1}^{\infty} \eta_k(t) U_{3k}(x) = \sum_{i=1}^p \sum_{k=1}^{\infty} \eta_k(t) U_{3k}(x) \\
&= \sum_{i=1}^p \sum_{k=1}^{\infty} \frac{3\pi_{12} R(R+2h^a)(h+h^a) \phi_i^a U_{3k}(x) e^{j(\omega_i t - \phi_i)}}{4\rho h N_k h^a \omega_k^2 \sqrt{\left[1 - (\omega_i / \omega_k)^2\right]^2 + 4\zeta_{mn}^2 (\omega_i / \omega_k)^2}} \\
&= \sum_{i=1}^p \left[-\frac{3\pi_{12} R(R+2h^a)(h+h^a) \phi_i^a U_{3k}(x) e^{j(\omega_i t - \phi_i)}}{4\rho h N_k h^a \omega_k^2 \sqrt{\left[1 - (\omega_i / \omega_k)^2\right]^2 + 4\zeta_{mn}^2 (\omega_i / \omega_k)^2}} \int_{x_i - L_a/2}^{x_i + L_a/2} \left[\frac{5R^3}{\left[(x-x_i)^2 + R^2\right]^{7/2}} \right. \right. \\
&\quad - \frac{4R}{\left[(x-x_i)^2 + R^2\right]^{5/2}} + \frac{5(2h^a + R)^3}{\left[(x-x_i)^2 + (2h^a + R)^2\right]^{7/2}} - \frac{4(2h^a + R)}{\left[(x-x_i)^2 + (2h^a + R)^2\right]^{5/2}} \\
&\quad \left. \left. - \frac{10(h^a + R)^3}{\left[(x-x_i)^2 + (h^a + R)^2\right]^{7/2}} + \frac{8(h^a + R)}{\left[(x-x_i)^2 + (h^a + R)^2\right]^{5/2}} \right] U_{3k} dx \right] \quad (22).
\end{aligned}$$

In the case of a cantilever beam, the maximum displacement occurs at the free end during vibration.

Therefore, it is essential to investigate the actuation effects on the displacement at the structural end.

2.3 Optimization based on the Newton-Raphson iteration method

The general format of an optimization iteration is as follows:

$$x_{k+1} = x_k - \Delta(x_k) \quad (23).$$

After derivation, in the context of the Newton-Raphson iteration method, z can be represented as:

$$z = -\left\{ \nabla f(x_k) [\nabla f(x_k)]^T \right\}^{-1} \nabla f(x_k) f(x_k) \quad (24).$$

in which $\nabla f(x_k) [\nabla f(x_k)]^T$ forms a square matrix. Consequently, the iterative expression for finding the root of a multivariate function can be obtained:

$$x_{k+1} = x_k - \left\{ \nabla f(x_k) [\nabla f(x_k)]^T \right\}^{-1} \nabla f(x_k) f(x_k) \quad (25).$$

Therefore, in order to determine the maximum or the optimal value of the function $f(x)$, it is only necessary to formulate the function as follows:

$$g(x) = f(x) - A \quad (26).$$

in which A represents the maximum value of the function $f(x)$ or the target-actuated value, the optimal solution can be obtained by substituting the objective function $g(x)$ into Eq.(25).

The terminal displacement u_3 of the cantilever beam is the scalar function to optimize. Thus, $u'_3 = u_3 - A$ represents the function $g(x)$, where u'_3 can be expressed as a multivariate function involving the positions of multiple actuators and the driving voltages. This can be mathematically expressed as:

$$u'_3 = u'_3(x_1, \varphi_1, x_2, \varphi_2, \dots, x_n, \varphi_n) \quad (27).$$

The column vector $\nabla u'_3$ can be represented as $\left(\frac{\partial u'_3}{\partial x_1}, \frac{\partial u'_3}{\partial \varphi_1}, \frac{\partial u'_3}{\partial x_2}, \frac{\partial u'_3}{\partial \varphi_2}, \dots, \frac{\partial u'_3}{\partial x_n}, \frac{\partial u'_3}{\partial \varphi_n} \right)$, the transpose of $\nabla u'_3$ resulting in a row vector. In order to ensure that the matrix $\nabla f(x_k) [\nabla f(x_k)]^T$ is invertible, it is necessary to make it positive definite. Therefore, the matrix $\nabla f(x_k) [\nabla f(x_k)]^T + I$ can be represented as:

$$\nabla f(x_k) [\nabla f(x_k)]^T + I = \begin{pmatrix} \frac{\partial u'_3}{\partial x_1} \cdot \frac{\partial u'_3}{\partial x_1} + 1 & \frac{\partial u'_3}{\partial x_1} \cdot \frac{\partial u'_3}{\partial \varphi_1} & \dots & \frac{\partial u'_3}{\partial x_1} \cdot \frac{\partial u'_3}{\partial \varphi_n} \\ \frac{\partial u'_3}{\partial \varphi_1} \cdot \frac{\partial u'_3}{\partial x_1} & \frac{\partial u'_3}{\partial \varphi_1} \cdot \frac{\partial u'_3}{\partial \varphi_1} + 1 & \dots & \frac{\partial u'_3}{\partial \varphi_1} \cdot \frac{\partial u'_3}{\partial \varphi_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial u'_3}{\partial \varphi_n} \cdot \frac{\partial u'_3}{\partial x_1} & \frac{\partial u'_3}{\partial \varphi_n} \cdot \frac{\partial u'_3}{\partial \varphi_1} & \dots & \frac{\partial u'_3}{\partial \varphi_n} \cdot \frac{\partial u'_3}{\partial \varphi_n} + 1 \end{pmatrix} \quad (28).$$

Substituting Eq.(28), u'_3 , and $\nabla f(x_k)$ into Eq.(25) yields the iterative formula for optimization:

$$\begin{pmatrix} x_1^{k+1} \\ \varphi_1^{k+1} \\ x_2^{k+1} \\ \varphi_2^{k+1} \\ \vdots \\ x_n^{k+1} \\ \varphi_n^{k+1} \end{pmatrix} = \begin{pmatrix} x_1^k \\ \varphi_1^k \\ x_2^k \\ \varphi_2^k \\ \vdots \\ x_n^k \\ \varphi_n^k \end{pmatrix} - \begin{pmatrix} \frac{\partial u'_3}{\partial x_1} \cdot \frac{\partial u'_3}{\partial x_1} + 1 & \frac{\partial u'_3}{\partial x_1} \cdot \frac{\partial u'_3}{\partial \varphi_1} & \dots & \frac{\partial u'_3}{\partial x_1} \cdot \frac{\partial u'_3}{\partial \varphi_n} \\ \frac{\partial u'_3}{\partial \varphi_1} \cdot \frac{\partial u'_3}{\partial x_1} & \frac{\partial u'_3}{\partial \varphi_1} \cdot \frac{\partial u'_3}{\partial \varphi_1} + 1 & \dots & \frac{\partial u'_3}{\partial \varphi_1} \cdot \frac{\partial u'_3}{\partial \varphi_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial u'_3}{\partial \varphi_n} \cdot \frac{\partial u'_3}{\partial x_1} & \frac{\partial u'_3}{\partial \varphi_n} \cdot \frac{\partial u'_3}{\partial \varphi_1} & \dots & \frac{\partial u'_3}{\partial \varphi_n} \cdot \frac{\partial u'_3}{\partial \varphi_n} + 1 \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial u'_3}{\partial x_1} \\ \frac{\partial u'_3}{\partial \varphi_1} \\ \vdots \\ \frac{\partial u'_3}{\partial \varphi_n} \end{pmatrix} \cdot (u_3 - A) \quad (29).$$

Considering the complexity of Eq.(22), which makes it inconvenient for differentiation, a numerical differentiation method is employed to approximate the first-order derivative of the function. For instance:

$$\begin{aligned}\frac{\partial u_3'}{\partial x_1} &\approx \frac{u_3'(x_1 + \Delta, \varphi_1, x_2, \varphi_2, \dots, x_n, \varphi_n) - u_3'(x_1, \varphi_1, x_2, \varphi_2, \dots, x_n, \varphi_n)}{\Delta} \\ \frac{\partial u_3'}{\partial \varphi_1} &\approx \frac{u_3'(x_1, \varphi_1, x_2, \varphi_2, \dots, x_n, \varphi_n) - u_3'(x_1, \varphi_1 + \Delta, x_2, \varphi_2, \dots, x_n, \varphi_n)}{\Delta}\end{aligned}\quad (30).$$

Taking into account that the noticeable range of electric field intensity gradient changes caused by the AFM probe is on the order of 10^{-8} , the value of Δ is chosen to be 10^{-10} , in which Δ represents the small increment in the independent variable. Therefore, it is evident that the Newton-Raphson iterative method is highly dependent on the choice of initial values.

3. Case Studies

As previously explained, the mitigation of stress concentration is sought through the strategic deployment of multiple actuators at diverse spatial locations. The parameters characterizing the elastic beam (constructed from polypropylene) and the actuator (utilizing BST technology) are delineated in Table 1. To illustrate the efficacy of the multifaceted actuation strategy is systematically examined and compared.

Table 1 Parameters and properties of the beam model

Property	Values
Beam length L , m	0.100
Beam width b , m	0.010
Beam thickness h , m	0.001
Young's modulus of beam Y , N/m ²	1.556×10^9
Beam mass density ρ , kg/m ³	907
Poisson's ratio μ	0.3
Flexoelectric patches thickness h^a , μm	50
Flexoelectric patches length L^a , m	0.010
Flexoelectric constant π_{12} , $\mu\text{C}/\text{m}$	100
AFM probe tip radius R , nm	50

Actuation voltage ϕ^a , V	100
Initial force F , N	0.05

3.1 Univariate analysis

3.1.1 Analysis of end displacement with respect to actuator position

In the context of the mode 1 as shown in **Fig.3(a)**, it is observed that the tip displacement of the cantilever beam exhibits a negative correlation with the positioning of the flexoelectric patch. This signifies that as the patch is situated in closer proximity to the fixed end of the beam, the tip displacement experiences an augmentation. The reason for this phenomenon is that, for the actuator, if only the position is changed while other parameters such as driving voltage remain unchanged, then according to Eq. (20), the resulting bending moment is a constant, rather than the applied force remaining unchanged. Therefore, when the applied moment is constant, the closer the load is to the fixed end, the greater the displacement at the free end. The gradient of the linear representation in the graphical depiction encapsulates the rate of alteration in the tip displacement concerning the patch position, manifesting as a consistently negative and nearly constant change rate.

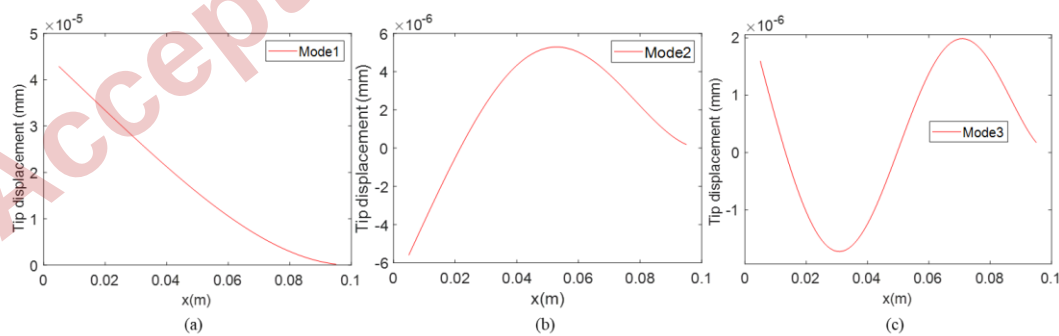


Fig. 3 Cantilever beam's end displacement of the response of the patch position; (a) mode 1 response of the patch position ; (b) mode 2 response of the patch position; (c) mode 3 response of the patch position;

In the realm of the mode 2 as shown in **Fig.3(b)**, a delicate equilibrium is discerned between the actuator position and the resulting end displacement. As the patch progressively distances itself from the

fixed end, the end displacement undergoes a transition from negative to positive, attaining its zenith before initiating a subsequent diminution. Within the purview of the mode 3 as shown in **Fig.3(c)**, an ostensibly sinusoidal relationship manifests between the actuator position and the consequential end displacement.

3.1.2 Analysis of end displacement with respect to actuated voltage

The provided **Fig.4(a)** to **(c)** illuminate a clear and direct correlation between the cantilever tip displacement and the applied actuating voltage. This signifies that the cantilever tip displacement exhibits a linear relationship, wherein changes in the actuated voltage directly influence the extent of displacement. This observed linearity underscores the responsiveness of the system to variations in the applied voltage.

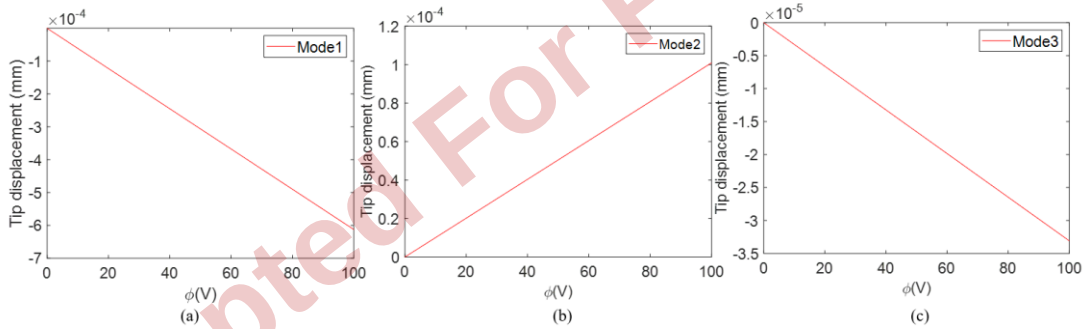


Fig. 4 Cantilever beam end displacement relative to excitation voltage modal response **(a)** mode 1 response of actuated voltage, **(b)** mode 2 response of actuated voltage, **(c)** mode 3 response of actuated voltage

To delve deeper into the comparative analysis of the three presented models, a horizontal examination of the graphs allows for a nuanced understanding of the flexoelectric effect's magnitude in each instance. Notably, the change rates of the lines in the graphs serve as indicative markers, representing the rate at which the cantilever tip displacement changes concerning alterations in the actuated voltage.

Upon scrutinizing these change rates, it becomes evident that the first model exhibits the steepest incline, signifying the highest sensitivity to changes in actuated voltage. Subsequently, the second model follows with a slightly less pronounced change rate, while the third model demonstrates the most gradual change in displacement concerning variations in the actuated voltage. This hierarchical arrangement of change rates across the models provides a quantitative insight into the relative strength of the flexoelectric effect within each system.

3.1.3 Analysis of end displacement with respect to flexoelectric patch thickness

In the subsequent triptych of **Fig.5(a)** to **(c)**, a consistent manifestation of patterns is observed in the reflections of the mode 1, mode 2, and mode 3. To illustrate, we examine the correlation between the terminal displacement of a cantilever beam and the flexoelectric sheet in the mode 1. Specifically, when the thickness of the flexoelectric sheet is below 1 micron, a notable escalation in the end displacement of the cantilever beam is discerned with increasing sheet thickness. However, once the thickness surpasses 1 micron, the end displacement undergoes negligible alteration with further increments in the thickness of the flexoelectric sheet. This comportment aligns with established theoretical frameworks[18].

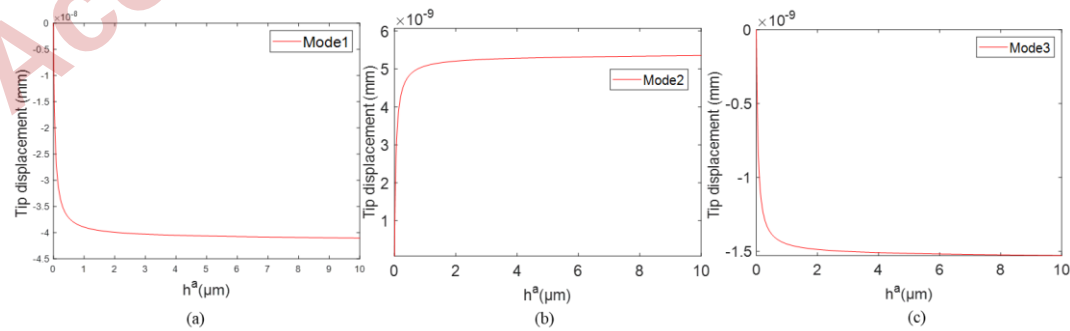


Fig.5 (a) mode 1 response of flexoelectric patch thickness, **(b)** mode 2 response of flexoelectric patch thickness, **(c)** mode 3 response of flexoelectric patch thickness,

Furthermore, it is pertinent to acknowledge the pragmatic constraints inherent in engineering applications, where the thickness of the flexoelectric sheet typically exceeds 10 microns. Consequently, we posit that, within the context of practical engineering endeavors, the impact of flexoelectric sheet thickness on terminal displacement may be deemed inconsequential.

3.1.4 Analysis of end displacement with respect to flexoelectric patch width

The above **Fig.6(a)** to **(c)** illustrates the influence of the flexoelectric patch width on the end displacement of the cantilever beam under different modal responses. The mode 1, mode 2, and mode 3 responses correspond to the fundamental, first harmonic, and second harmonic frequencies of the beam vibration, respectively. The **Fig.6(a)** to **(c)** shows that the end displacement of the beam is almost constant for different values of the flexoelectric patch width L^a , ranging from 0 to 100 mm. The change rate of the displacement versus L^a curve is close to zero, indicating that the flexoelectric patch width has negligible effect on the beam deflection.

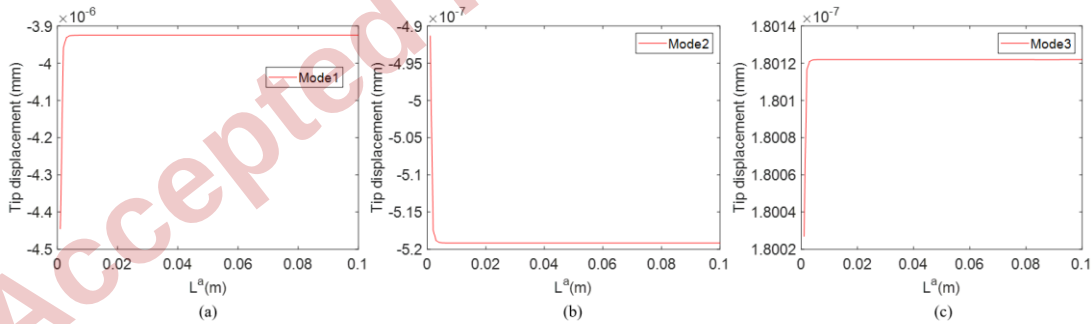


Fig.6 (a) mode 1 response of the flexoelectric patch width, **(b)** mode 2 response of the flexoelectric patch width, **(c)** mode 3 response of the flexoelectric patch width

This can be explained by the fact that the non-uniform electric field applied by the AFM probe is confined to a nanoscale region near the tip of the probe, where the electric field gradient is significant. The electric field gradient induces a flexoelectric polarization in the dielectric material, which in turn generates a flexoelectric force that acts on the beam. However, the flexoelectric force is only effective

within a small interval of no more than 10 nm in the core region of the beam, where the strain gradient is large. In other regions of the beam, the electric field intensity and the strain gradient are almost zero, and thus the flexoelectric force is negligible. Therefore, the end displacement of the beam is mainly determined by the applied load and the beam properties, and is insensitive to the width of the flexoelectric patch.

3.2 Optimal actuated position analysis

The calculation method employed in this study utilizes the Newton-Raphson iteration method, which is an adaptive gradient iterative approach capable of rapidly solving pole problems for complex multivariate functions. We chose the free end of the cantilever beam as the control target, with the goal of maximizing its driven displacement. To obtain the optimal sensor positions for actuation, it is necessary to initially roughly identify potential combinations of sensor positions before the computation, ensuring non-overlapping placement of sensors. The Newton-Raphson iteration method, employing the voltages of multiple sensors as input variables, can rapidly compute the optimal combinations of these positions. The optimal sensor positions can then be selected from the output results.

Below are the optimal results filtered by the Newton-Raphson iteration method, along with the displacement distribution figures resulting from the action of different numbers of actuators in the cantilever beam drive.

Table 2 The observed optimal positions actuated by two actuators

Mode number	Actuator 1, m	Actuator 2, m
1	0.00500	0.01500
2	0.047997	0.057997
3	0.00500	0.07100

Table 3 The observed optimal positions actuated by three actuators

Mode number	Actuator 1, m	Actuator 2, m	Actuator 3, m
-------------	---------------	---------------	---------------

1	0.00500	0.01500	0.02500
2	0.04300	0.05488	0.06663
3	0.00500	0.06573	0.07573

Table 4 The observed optimal positions actuated by four actuators

Mode number	Actuator 1, m	Actuator 2, m	Actuator 3, m	Actuator 4, m
1	0.00500	0.01500	0.02500	0.03500
2	0.03800	0.04871	0.05918	0.06918
3	0.00500	0.06100	0.07110	0.08101

Table 5 The observed optimal positions actuated by five actuators

Mode number	Actuator 1, m	Actuator 2, m	Actuator 3, m	Actuator 4, m	Actuator 5 m
1	0.005	0.015	0.025	0.035	0.045
2	0.03301	0.04301	0.05301	0.06301	0.07301
3	0.00500	0.05613	0.06613	0.07613	0.08613

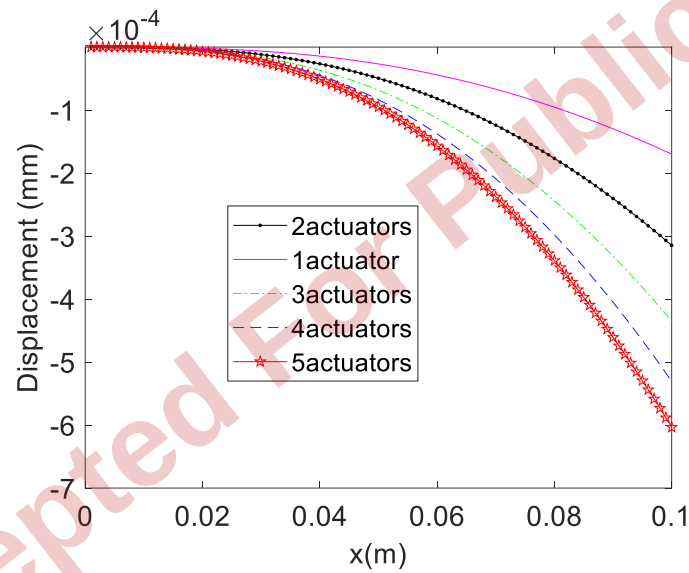


Fig. 7 The displacement caused by different numbers of actuators on the first mode of a cantilever beam

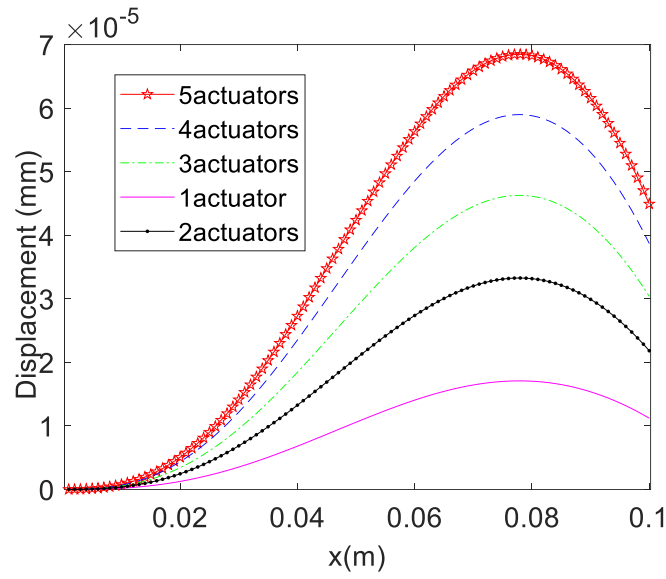


Fig. 8 The displacement caused by different numbers of actuators on the second mode of a cantilever beam

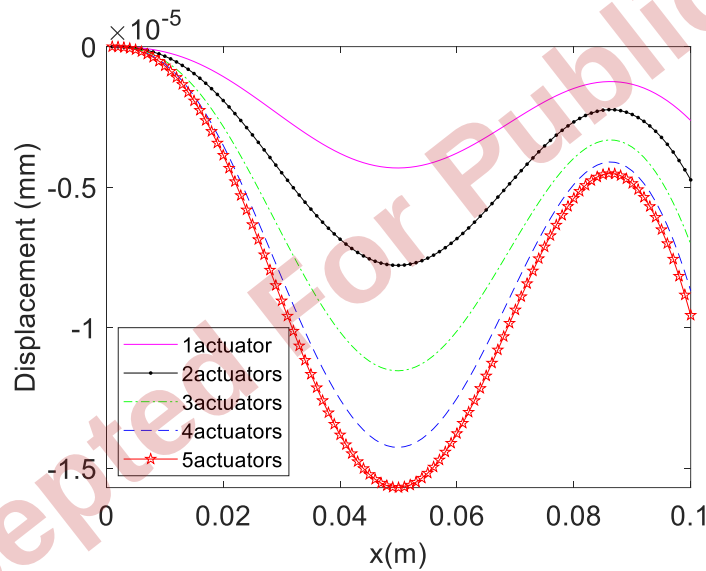


Fig. 9 The displacement caused by different numbers of actuators on the third mode of a cantilever beam

From **Fig.7-9**, it is evident that with an increasing number of actuators, the displacement generated by the actuators becomes larger. It is also significant that adding actuators is allowing for a greater range of displacement adjustment, which is more conducive to driving or controlling under different conditions. Additionally, considering that stress is directly proportional to voltage and also positively correlated with displacement, when the same displacement is generated, the voltage required for multiple actuators becomes smaller, which can greatly alleviate the situation of stress concentration. It is worth noting that

the x-axis variables in Figures 7 to 9 have the same meaning as the variable in Figure 3, all representing the position distribution along the cantilever beam.

The effectiveness of the Newton-Raphson iteration method in optimizing sensor positions is demonstrated by the significant improvements in displacement control and stress distribution. By ensuring optimal placement and reducing voltage requirements, this method enhances the overall performance and longevity of the cantilever beam system.

3.3 Optimal control position analysis

The Newton-Raphson iteration method can solve the zero-point problems of complex multivariate functions, allowing for the calculation of the optimal optimized positions of sensors. Similarly, we chose the free end of the cantilever beam as the control target, with the goal of achieving zero displacement. Using the positions and voltages of multiple sensors as input variables, it can rapidly compute the optimal combinations of these positions. Then, the optimal sensor positions can be selected from the output results.

Table 6 The observed optimal positions controlled by three actuators

Mode number	Actuator 1; m,V	Actuator 2; m,V	Actuator 3; m,V
2	0.0150, 0.4997	0.02500,0.5000	0.0860,0.5000
3	0.0200,2.0013	0.04091,3.0008	0.0600,4.0000

Table 7 The observed optimal positions controlled by four actuators

Mode number	Actuator1;m,V	Actuator2;m,V	Actuator3;m,V	Actuator4;m,V
2	0.011644,1.006101	0.021644,99.95029	0.036078,1.000	0.095, 1.00560
3	0.015, 9.91637653	0.07048, 0.006018	0.08048,9.9902	0.005, 9.98658

From the above two tables, we successively obtained the distribution results of the optimal control positions for three and four actuators. Given the large number of different position combinations for multiple sensors, traditional exhaustive methods suffer from low computational efficiency[18]. In

contrast, the Newton-Raphson iteration method effectively addresses this issue. Prior to the calculations, it is only necessary to identify potential optimal positions, and upon substitution, accurate results can be swiftly obtained.

By strategically arranging sensor positions, the signal from sensors can be significantly enhanced, enabling precise monitoring of structural vibrations dominated by bending deformations. This precise monitoring is crucial for maintaining structural integrity and ensuring the longevity of the system. The enhanced signal-to-noise ratio achieved through optimal sensor placement allows for more accurate detection and analysis of vibrational modes, leading to better control strategies and improved performance of the cantilever beam system.

3.4 Negative feedback control of cantilever beam step response

A new flexoelectric beam vibration control method based on the Newton-Raphson optimization algorithm is proposed. In the previous analysis, both the position of the flexoelectric beam and the magnitude of the driving voltage significantly affect the tip displacement of the cantilever beam. Therefore, in this study, based on the previously calculated data, the corresponding optimal position for placing the flexoelectric beam is selected when analyzing different modes. Real-time amplitude feedback is used to control the driving voltage to achieve good end displacement control effects. During the feedback process, the voltage of the flexoelectric beam is optimized as the optimization variable, and the current most suitable voltage amplitude is calculated using the Newton-Raphson method.

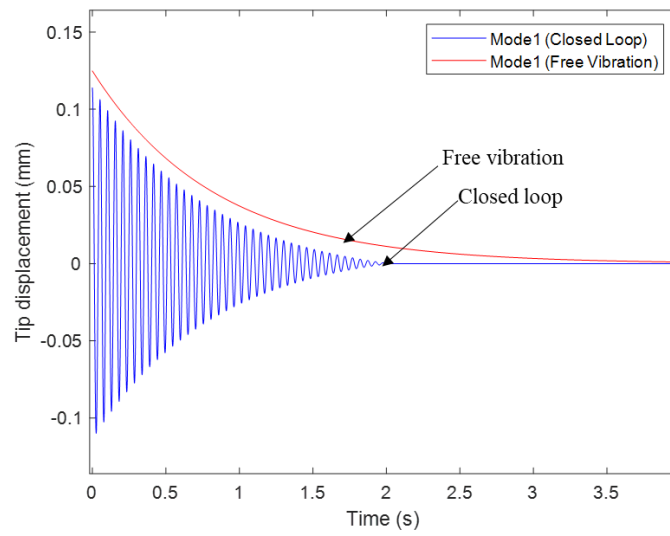


Fig. 10 The actual oscillation line and natural decay envelope line of the mode 1 of a cantilever beam after being controlled by the Newton iteration method

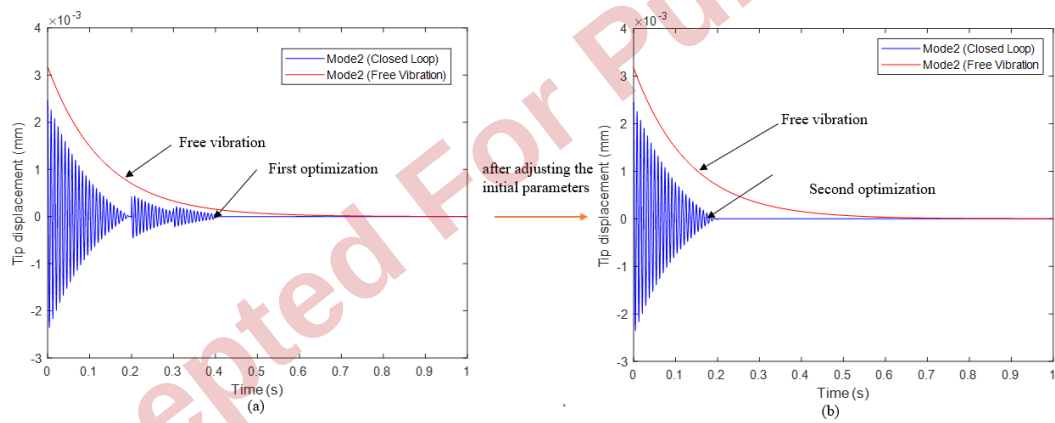


Fig. 11 (a) The actual oscillation line and natural decay envelope line of the mode 2 of a cantilever beam after initial control; (b) The oscillation line of a cantilever beam's mode 2 after optimized control

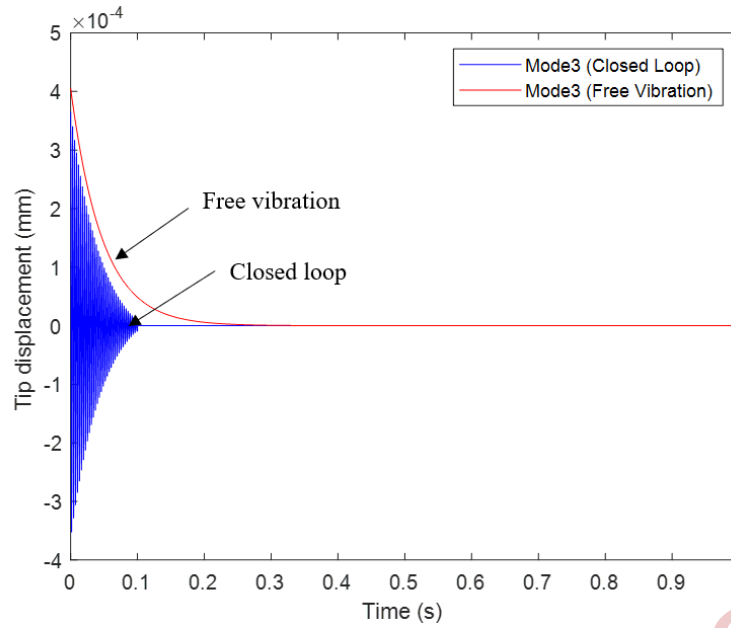


Fig. 12 The actual oscillation line and natural decay envelope line of the mode 3 of a cantilever beam after control

As shown in **Fig.10-12**, the four flexoelectric actuators effectively control the cantilever beam vibration. In **Fig.10**, after optimizing control for the mode 3, stable conditions are restored in only about 2 seconds compared to before optimization, which is approximately a 50% reduction in time. The situation is slightly different for the mode 2. In **Fig.11(a)**, the optimization control interval is 0.1 seconds, and the same initial value is selected for each optimization. As a result, the phenomenon of repeated fluctuations occurs after 0.2 seconds. The reason for this is that the amplitude decreases significantly compared to before 0.2 seconds, causing reverse driving due to the Newton-Raphson optimization initial value being determined by the size of the initial value. After considering this factor, adjusting the iteration initial value after 0.2 seconds results in **Fig.11(b)**, in which stability is basically achieved at around 0.2 seconds, which greatly improves compared to the previous 0.8 seconds. Similarly, in the mode 3, from **Fig.12**, selecting a smaller iteration initial value after 0.1 seconds will not result in reverse driving, and the control effect is also significant.

4. Conclusions

Due to stress concentration induced by flexoelectric actuators, placing multiple actuators at optimal positions can enhance both the driving performance and control reliability of the flexoelectric control system. Atomic Force Microscope (AFM) probes were employed to induce changes in the electric field gradient, leveraging the converse flexoelectric effect to generate concentrated stress in the flexible actuator. Optimal positions for actuators in modes 1, 2, and 3 were explored to achieve optimal control effects. Initially, an analysis was conducted on the impact of variables such as the thickness of the flexoelectric patch, its position, width, and driving voltage on the end displacement of the cantilever beam under the condition of controlling a single variable. Based on the findings of this study, considering the limited influence of flexoelectric patch thickness and width on the end displacement of the cantilever beam, a Newton-Raphson iteration method was introduced to rapidly compute the optimal distribution positions of actuators for driving and vibration control, along with their corresponding driving voltages. The Newton-Raphson iteration method relies on the selection of initial values, so after roughly identifying possible optimal scenarios, substituting them into the calculations yields rapid and accurate results. The computational results demonstrate that, compared to traditional exhaustive algorithms, the Newton-Raphson iteration method requires less time for calculating optimal solutions and is more accurate, providing a highly efficient means to obtain precise results. In addition, for the end displacement control of a flexoelectric cantilever beam, compared to the traditional open-loop control method, feedback control based on the Newton-Raphson iteration method can quickly converge the end displacement of the cantilever beam to 0. The computational results also indicate that selecting different initial values for iteration at different time periods can yield better results.

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